Designing a Nearly Optimal Quantum Algorithm for Linear Differential Equations via Lindbladians

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Solving linear ordinary differential equations (ODEs) is one of the most promising applications for quantum computers to demonstrate exponential advantages. The challenge of designing a quantum ODE algorithm is how to embed nonunitary dynamics into intrinsically unitary quantum circuits. In this Letter, we propose a new quantum algorithm for solving ODEs by harnessing open quantum systems. Specifically, we propose a novel technique called nondiagonal density matrix encoding, which leverages the inherent nonunitary dynamics of Lindbladians to encode general linear ODEs into the nondiagonal blocks of density matrices. This framework enables us to design quantum algorithms with both theoretical simplicity and high performance. Combined with the state-of-the-art quantum Lindbladian simulation algorithms, our algorithm can outperform all existing quantum ODE algorithms and achieve near-optimal dependence on all parameters under a plausible input model. We also give applications of our algorithm including the Gibbs state preparations and the partition function estimations.

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Introduction—Differential equations [1] have long served as an essential tool for modeling and describing the dynamics of systems in both natural and social sciences. A general linear ordinary differential equation (ODE) can be written as

$$\frac{d}{dt}\vec{\mu}(t) = -V(t)\vec{\mu}(t) + \vec{b}(t), \tag{1}$$

where $V(t) \in \mathbb{C}^{2^n \times 2^n}$, $\vec{\mu}(t), \vec{b}(t) \in \mathbb{C}^{2^n}$, and the initial condition is given by $\vec{\mu}(0) = \vec{\mu}_0$. Classical simulation algorithms often become highly inefficient for large systems due to their polynomial scaling with respect to the system dimension. In contrast, quantum algorithms [2–11] with appropriate input access can generate a quantum state encoding the solution of the ODE with only polylogarithmic dependence on the system dimension. This remarkable efficiency makes solving ODE a promising application of quantum computers. In particular, the Hamiltonian simulation [12–17], which aims to simulate the Schrödinger equation, a special case of ODEs, on quantum computers, is arguably one of the most significant applications and may be among the first to demonstrate practical quantum advantages.

Since $V(t)$ is generally not anti-Hermitian, the time evolution operator for the ODE is not necessarily unitary. This is the case in many important problems, such as those arising in non-Hermitian physics [18–31,90] and fluid dynamics [32–37]. The central challenge in designing quantum algorithms for linear ODEs is how to embed nonunitary dynamics into intrinsically unitary quantum dynamics. When $V(t)$ is a time-independent normal matrix and $\vec{b}(t) = \vec{0}$, the problem of solving an ODE can be efficiently addressed by the powerful quantum singular value transformation (QSVT) algorithm [38]. However, for ODEs with a time-dependent non-normal matrix $V(t)$ and a possibly inhomogeneous term $\vec{b}(t) \neq \vec{0}$, QSVT is no longer applicable due to the mismatch between the singular value transformation and the eigenvalue transformation.

Previous works have developed two main strategies for solving general linear ODEs. The first is the linear-system-based approach, which discretizes the ODE by a numerical scheme, reformulates the discretized ODE as a dilated linear system of equations and solves the linear system by quantum linear system algorithms. The first efficient linear-system-based approach was proposed by Berry in [2], where multistep methods for time discretization and the Harrow-Hassidim-Lloyd algorithm [39] were applied to solving the resulting linear system. Since then, several subsequent works [3–6] have advanced the

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linear-system-based approach by employing higher-order time discretization and more advanced quantum linear system algorithms [40–46]. The second strategy is the evolution-based approach, which directly embeds the time evolution operator into the subspace of an efficiently implementable unitary by time marching [7], reducing to Hamiltonian simulation problems [8–10], or quantum eigenvalue processing [11].

Beyond these methodologies, it remains an intriguing challenge to develop quantum ODE algorithms with improved efficiency by reducing the ODE to other quantumly solvable tasks. For instance, it is rather tempting to use natural nonunitary dynamics of open quantum systems to solve nonunitary ODEs, such as the Lindblad master equation (Lindbladian) [47]. Unlike previous evolution-based algorithms where nonunitary dynamics is embedded into the subspace of unitaries (block encoding [17]), dynamics of subsystems described by Lindbladians are naturally nonunitary due to the interaction with the environment. The Lindbladian itself has been proven to be a universal quantum computing model [48,49]. Recently, there have been several works proposing quantum algorithms based on Lindbladians, where they aim to encode states [50] of interest such as Gibbs states [51–54] and ground states [55,56] as the steady states of Lindbladians. Inspired by these rapid developments, our question arises: Is it possible to solve linear ODEs via Lindbladians? There are already various quantum algorithms for simulating Lindbladians [57–64], which makes them rather amenable for solving linear ODEs via Lindbladians on quantum computers as long as we can establish a connection between them.

In this Letter, we provide a positive answer to the above question. We demonstrate how general linear ODEs can be embedded into Lindbladians with the aid of a new technique called nondiagonal density matrix encoding. Based on this result, we develop an efficient quantum algorithm for solving ODEs by leveraging state-of-the-art Lindbladian simulation quantum algorithms [61,63] while considering two different input models for the coefficient matrix $V(t)$. Notably, under the second input model, our algorithm outperforms all existing quantum algorithms and achieves near-optimal dependence on all parameters. We note that Ref. [65] also explores connections between ODEs and Lindbladians, using different techniques and mappings to show that any ODE can be associated with a Markovian quantum master equation and providing conditions under which the master equation is Lindbladian. Therefore, the novelty of our Letter lies mainly in achieving a computational breakthrough, rather than in the observation of representability.

Algorithm overview—If a system is coupled to an environment that is sufficiently large for the Markovian approximation to hold, the dynamics of the system can be modeled by the Lindbladian [47]

$$\frac{d\rho}{dt} = \mathcal{L}[\rho] = -i[H(t),\rho] + \sum_i \left(F_i(t)\rho F_i(t)^\dagger - \frac{1}{2}\{\rho, F_i(t)^\dagger F_i(t)\} \right), \quad (2)$$

where $\rho = \sum_{ij} \rho_{ij} |i\rangle\langle j|$ is the density matrix of the system, $H(t)$ is the internal Hamiltonian, and $\{F_i(t)\}$ are quantum jump operators. By treating ρ as $\vec{u}(t)$ in an ODE, the Lindbladian naturally corresponds to an ODE with a generally non-normal $V(t)$. However, the Lindbladian has to be a completely positive trace-preserving map [66] and cannot be used to encode general ODE problems.

To address this issue, we propose a technique called nondiagonal density matrix encoding (NDME), which was introduced in Ref. [67]. Similar to the concept of block encoding, where a matrix M of interest is encoded into the upper-right block of a unitary operator [17] [also see the definition in Sec. A2 of Supplemental Material (SM) [68]], NDME aims to encode the matrix M in a nondiagonal block of a density matrix to jump out of the restrictions of Hermiticity and positive semidefiniteness.

Definition 1: NDME—Given an $(l+n)$-qubit density matrix ρ_M and an n-qubit matrix M, if ρ_M satisfies $(\langle s_1|_l \otimes I_n)\rho_M(|s_2\rangle_l \otimes I_n) = \gamma M$, where $|s_1\rangle_l$ and $|s_2\rangle_l$ are two different computational basis states of the l-qubit ancilla system and $\gamma \geq 0$, then ρ_M is called an $(l+n, |s_1\rangle, |s_2\rangle, \gamma)$ NDME of M.

In this Letter, we focus mainly on a $(1+n, |0\rangle, |1\rangle, \gamma)$ NDME where ρ_M has the form $\rho_M = \begin{pmatrix} \cdot & \gamma M \\ \gamma M^\dagger & \cdot \end{pmatrix}$. Since ρ_M is Hermitian, it is also a $(1+n, |1\rangle, |0\rangle, \gamma)$ NDME of $M^\dagger$.

We now show how to combine Lindbladians and NDME to encode general linear ODEs. Starting with a $(1+n)$-qubit initial state $\rho_0 = |+\rangle\langle+| \otimes |\mu_0\rangle\langle\mu_0|$ which is a $(1+n, |0\rangle, |1\rangle, \frac{1}{2})$ NDME of $|\mu_0\rangle\langle\mu_0|$, we consider a Lindbladian having the form of Eq. (2) with $H(t) = \begin{pmatrix} H_1(t) & 0 \\ 0 & \alpha I_n \end{pmatrix}$ and $F_i(t) = \begin{pmatrix} G_i(t) & 0 \\ 0 & \beta_i I_n \end{pmatrix}$. A focus on the upper-right block of ρ_0, $\mathcal{L}[\rho_0]$ gives

$$\frac{1}{2}|\mu_0\rangle\langle\mu_0| \to -i[H_1(t)-\alpha]\frac{1}{2}|\mu_0\rangle\langle\mu_0| - \left(\frac{1}{2}\sum_i G_i(t)^\dagger G_i(t) - \sum_i \beta_i^* G_i(t) + \frac{1}{2}\sum_i |\beta_i|^2\right)\frac{1}{2}|\mu_0\rangle\langle\mu_0|. \quad (3)$$

Here, since the α term can be absorbed into $H_1(t)$ and $\frac{1}{2}\sum_i G_i(t)^\dagger G_i(t)$ is positive semidefinite, for semidissipative ODEs [the Hermitian part of $V(t)$ is positive semidefinite], we can simply set $\alpha = \beta_i = 0$ such that, on the left vector of this upper-right block, we realize the homogeneous linear ODE

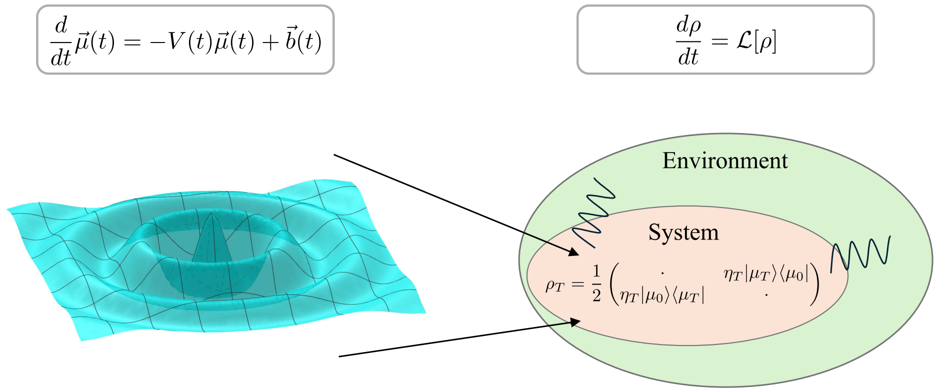

FIG. 1. High-level illustration of the algorithm. The dynamics of a linear ODE is embedded into the intrinsic nonunitary dynamics of a Lindbladian, and the solution to the ODE is encoded in a nondiagonal block of the density matrix evolved according to the Lindbladian.

$$\frac{d\vec{\mu}(t)}{dt} = -V(t)\vec{\mu}(t) = \left(-iH_1(t) - \frac{1}{2}\sum_i G_i(t)^\dagger G_i(t)\right)\vec{\mu}(t), \tag{4}$$

with $\vec{\mu}(0) = |\mu_0\rangle$. Equation (4) shows the interesting but expected connection between the anti-Hermitian (oscillating) part of $V(t)$ and the internal Hamiltonian of the Lindbladian and the connection between the Hermitian (dissipative) part of $V(t)$ and the environment-induced jump operators of the Lindbladian. Evolving the Lindbladian for a time T, we will have

$$\rho_T = \frac{1}{2}\begin{pmatrix} \cdot & \eta_T|\mu_T\rangle\langle\mu_0| \\ \eta_T|\mu_0\rangle\langle\mu_T| & \cdot \end{pmatrix}, \tag{5}$$

where $|\mu_T\rangle = \vec{\mu}(T)/\eta_T$ is the normalized solution with $\eta_T = \|\vec{\mu}(T)\|$. A high-level picture of these procedures is shown in Fig. 1. More details can be found in Sec. B of SM [68].

In the above discussions, we noted that it is possible to set $\beta_i \neq 0$ for nondissipative ODEs. However, we will focus on the semidissipative case because it has been shown that there is a no-go theorem with exponential worst-case complexity for the nondissipative case [80], which states that a generic quantum algorithm for ODEs with growing components must exhibit a worst-case complexity of $e^{\Omega(T)}$. Such exponential scaling is considered inefficient. Consequently, existing quantum algorithms for ODEs are typically designed for the semidissipative case, and our Letter follows this established convention.

Having prepared ρ_T, we can further extract the information of $\vec{\mu}(T)$ depending on our purposes. Since the information of $\vec{\mu}(T)$ has already been encoded in ρ_T, we can perform measurements on ρ_T to acquire properties of $\vec{\mu}(T)$ such as $\eta_T\langle\mu_0|\mu_T\rangle$ known as the Loschmidt echo [81] and the expectation value $\eta_T^2\langle\mu_T|O|\mu_T\rangle$ with respect to some observable O. We give concrete measurement strategies in Sec. B3 of SM [68]. With the aid of amplitude estimation algorithms [82–84], we can use $\mathcal{O}(\eta_T^{-1}\epsilon^{-1})$ queries to the preparation oracle of $|S_T\rangle$, which is the purification of ρ_T, to give an $\eta_T\epsilon$ additive estimation of $\eta_T\langle\mu_0|\mu_T\rangle$, and we can use $\mathcal{O}(\eta_T^{-2}\epsilon^{-1})$ queries to the preparation unitary of $|S_T\rangle$ to give an $\eta_T^2\epsilon$ additive estimation of $\eta_T^2\langle\mu_T|O|\mu_T\rangle$. We emphasize that the purification state $|S_T\rangle$ can be directly obtained from the Lindbladian simulation algorithms [61,63] that we adopt in this Letter. Even if we have access only to ρ_T, we can still measure these properties via Hadamard tests [85] with a standard quantum limit scaling (ϵ^{-2}).

If our purpose is to prepare $|\mu_T\rangle$ rather than merely measuring properties, which is also the setting in other quantum algorithms for ODEs, we can instead use the amplitude amplification algorithm [86] to extract $|\mu_T\rangle$ from $|S_T\rangle$. The observation is that we have the relation $(\langle 0| \otimes I_n \otimes \langle E_{T1}|_e)|S_T\rangle = \eta_T/\sqrt{2}|\mu_T\rangle$, where $|E_{T1}\rangle_e$ is a special environment state, which tells one how to design the Grover operator [87] for the amplitude amplification. As a result, to prepare $|\mu_T\rangle$, we require $\mathcal{O}(\eta_T^{-1})$ queries on the Lindbladian simulation oracle and the initial state ($|\mu_0\rangle$) preparation oracle which matches the lower bound [80]. See Sec. B4 of SM [68] for detailed constructions.

Complexity analysis—In this section, we discuss the overall complexity of our algorithm. To facilitate the comparison with previous algorithms, we focus on the preparation of $|\mu_T\rangle$ as the goal. Given an ODE in Eq. (1) with $A(t) = [V(t) - V^\dagger(t)]/2i = H_1(t)$ and $B(t) = [V(t) + V^\dagger(t)]/2 = (1/2)\sum_i G_i(t)^\dagger G_i(t)$, the complexity now depends on two parts: input models and the Lindbladian simulation algorithms. For the Lindbladian simulation, we adopt the results in Ref. [61] for the time-independent case and the results in Ref. [63] for the time-dependent case. To enable the Lindbladian simulation, we need access (block encodings) of $H_1(t)$ and $\{G_i(t)\}$, which naturally leads to two different input models.

The first model is that we are given the direct access (block encoding) of $V(t)$. In this case, we need to construct the block encodings of $H_1(t)$ and $\{G_i(t)\}$ from $V(t)$.

For $\{G_i(t)\}$, we consider setting a single jump operator, $G(t) = \sqrt{2B(t)}$, constructed by implementing a modified square root function approximated by QSVT [38,88] using only odd polynomials (see Sec. B5 of SM [68]), which introduces some overheads in terms of α_V and ϵ and also gives an additional complexity dependence on Δ, the lower bound of the smallest nonzero eigenvalues of $B(t)$ [89]. The second input model is the square root access model where we are given the block encoding of $H_1(t)$ and $\{G_i(t)\}$ in the form $\alpha_{H_1}^{-1}|0\rangle\langle 0| \otimes H_1(t) + \sum_i \alpha_{G_i}^{-1}|i\rangle\langle i| \otimes G_i(t)$. This is the most natural model for Lindbladian simulations and directly applies to several practical scenarios where the operator can be expressed in a sum-of-squares form. For example, consider the case where $B = \sum_{ij} h_{ij} c_i^\dagger c_j$ is a free fermionic Hamiltonian with the matrix $h = \sum_{ij} h_{ij}|i\rangle\langle j|$. In such cases, it is always possible to find a basis transformation T such that $h = T^\dagger \Sigma T$ with a diagonal Σ. Expressing B in terms of the transformed creation and annihilation operators $c_i'^\dagger$ and c_i' with $c_i' = \sum_j T_{ij} c_j$ satisfies the square root access. For more general cases, see Refs. [90–92].

Depending on the input models, we have either of the following two theorems.

Theorem 1: Quantum ODE solver via Lindbladian, direct access model—For semidissipative and homogeneous linear ODEs, assume we are given access to α_V-block encoding $U_{V(t)}$ of $V(t)$ with $\alpha_V \geq \max_t \|V(t)\|$ and quantum state preparation unitary $U_{\mu_0}\colon |0\rangle_n \to |\mu_0\rangle$ and suppose that the smallest nonzero eigenvalues of $B(t)$ is lower bounded by $\Delta > 0$. Then there exists a quantum algorithm that outputs an ϵ approximation of $|\mu_T\rangle$ by using $\mathcal{O}(\eta_T^{-1})$ queries to U_{μ_0} (and its reverse) and $\tilde{\mathcal{O}}\{\eta_T^{-1}\Delta^{-1}\alpha_V^2 T[\log^2(1/\epsilon)/\log\log(1/\epsilon)]\}$ queries to $U_{V(t)}$ (and its inverse) in the time-independent case, or $\tilde{\mathcal{O}}(\eta_T^{-1}\Delta^{-1}\alpha_V^2 T\{\eta_T^{-1}\Delta^{-1}\alpha_V^2 T[\log^3(1/\epsilon)/\log^2\log(1/\epsilon)]\})$ queries to $U_{V(t)}$ (and its inverse) in the time-dependent case.

Theorem 2: Quantum ODE solver via Lindbladian, square root access model—For semidissipative and homogeneous linear ODEs, assume that we are given access to one-block encoding $U_{H_1(t),G_i(t)}$ of $\alpha_V^{-1}|0\rangle\langle 0| \otimes H_1(t) + \sum_i \alpha_V^{-1/2}|i\rangle\langle i| \otimes G_i(t)$, with $H_1(t) = A(t)$, $\sum_i G_i(t)^\dagger G_i(t) = B(t)$, and $\alpha_V \geq \max_t \|V(t)\|$, and access to the quantum state preparation unitary $U_{\mu_0}\colon |0\rangle_n \to |\mu_0\rangle$. Then there exists a quantum algorithm that outputs an ϵ approximation of $|\mu_T\rangle$ by using $\mathcal{O}(\eta_T^{-1})$ queries to U_{μ_0} (and its reverse) and $\tilde{\mathcal{O}}\{\eta_T^{-1}\alpha_V T[\log(1/\epsilon)/\log\log(1/\epsilon)]\}$ queries to $U_{H_1(t),G_i(t)}$

TABLE I. Comparison of various ODE solvers for homogeneous cases under two different input models, the direct access model (Theorem 1) and the square root access model (Theorem 2). The symbol ϵ is defined as the preparation error of the normalized state $|\mu_T\rangle$. We define $\eta_T = \|\vec{\mu}(T)\|$ with $\vec{\mu}(0) = |\mu_0\rangle$. Δ is defined as the lower bound of the smallest nonzero eigenvalues of the Hermitian part of $V(t)$. κ_V is the upper bound of condition numbers of $V(t)$.

Algorithms	Direct access model queries to $V(t)$	Square root access model queries to $H_1(t)$ and $\{G_i(t)\}$	Queries to $\lvert\mu_0\rangle$
Spectral method [4]	$\tilde{\mathcal{O}}\Big\{\eta_T^{-1}\kappa_V\alpha_V T\mathrm{poly}[\log(1/\epsilon)]\Big\}$	No change	$\tilde{\mathcal{O}}\Big\{\eta_T^{-1}\kappa_V\alpha_V T\mathrm{poly}[\log(1/\epsilon)]\Big\}$
Truncated Dyson [6]	$\tilde{\mathcal{O}}\Big(\eta_T^{-1}\alpha_V T[\log(1/\epsilon)]^2\Big)$	No change	$\mathcal{O}\Big[\eta_T^{-1}\alpha_V T\log(1/\epsilon)\Big]$
Quantum eigenvalue transformation (QEVT) (only time independent) [11]	$\tilde{\mathcal{O}}\Big\{\eta_T^{-1}[\alpha_V T + \log(1/\epsilon)]\log(1/\epsilon)\Big\}$ [96]	No change	$\tilde{\mathcal{O}}\Big\{\eta_T^{-1}[\alpha_V T + \log(1/\epsilon)]\log(1/\epsilon)\Big\}$
Time marching [7]	$\tilde{\mathcal{O}}\Big[\eta_T^{-1}\alpha_V^2 T^2\log(1/\epsilon)\Big]$	No change	$\mathcal{O}(\eta_T^{-1})$
LCHS [8]	$\tilde{\mathcal{O}}(\eta_T^{-2}\alpha_V T\epsilon^{-1})$	No change	$\mathcal{O}(\eta_T^{-1})$
Improved LCHS (time independent) [10]	$\tilde{\mathcal{O}}\Big(\eta_T^{-1}\alpha_V T[\log(1/\epsilon)]^{1+o(1)}\Big)$	No change	$\mathcal{O}(\eta_T^{-1})$
Improved LCHS (time dependent) [10]	$\tilde{\mathcal{O}}\Big(\eta_T^{-1}\alpha_V T[\log(1/\epsilon)]^{2+o(1)}\Big)$	No change	$\mathcal{O}(\eta_T^{-1})$
This Letter (time independent)	$\tilde{\mathcal{O}}\Big(\eta_T^{-1}\Delta^{-1}\alpha_V^2 T[\log^2(1/\epsilon)/\log\log(1/\epsilon)]\Big)$	$\tilde{\mathcal{O}}\Big(\eta_T^{-1}\alpha_V T[\log(1/\epsilon)/\log\log(1/\epsilon)]\Big)$	$\mathcal{O}(\eta_T^{-1})$
This Letter (time dependent)	$\tilde{\mathcal{O}}\Big(\eta_T^{-1}\Delta^{-1}\alpha_V^2 T[\log^3(1/\epsilon)/\log^2\log(1/\epsilon)]\Big)$	$\tilde{\mathcal{O}}\Big(\eta_T^{-1}\alpha_V T[\log^2(1/\epsilon)/\log^2\log(1/\epsilon)]\Big)$	$\mathcal{O}(\eta_T^{-1})$

(and its inverse) in the time-independent case, or $\tilde{\mathcal{O}}(\eta_T^{-1}\alpha_V T[\log^2(1/\epsilon)/\log^2\log(1/\epsilon)])$ queries to $U_{H_1(t),G_i(t)}$ (and its inverse) in the time-dependent case.

In the descriptions of Theorems 1 and 2, the block encoding model for time-dependent cases follows the Time-dependent Hamiltonian block encoding in Ref. [93] with an additional time register. (For more details, see Sec. B5 in SM [68], where we also explicitly provide additional algorithmic costs, including gate counts and the number of required ancilla qubits).

We summarize the complexity of our algorithm with a comparison to the previous ones in Table I. The lower bound of solving the ODE with respect to T is $\mathcal{O}(T)$ from the no fast-forwarding theorem [94], with respect to η_T is $\mathcal{O}(\eta_T^{-1})$ originating from the state discrimination [80], and with respect to ϵ is $\mathcal{O}[\log(1/\epsilon)/\log\log(1/\epsilon)]$ from the parity checking [95]. Under the direct access model, our algorithm gives near-optimal dependence on T, η_T, and ϵ. Under the square root access model, for previous methods we need to reconstruct the block encoding of $V(t)$ from $U_{H_1(t),G_i(t)}$. Our algorithm outperforms all previous methods under this model. Compared to methods requiring quantum linear system solvers [4,6,11], our algorithm achieves optimal queries to the initial state preparation. Compared to the time-marching method [7], our algorithm improves the dependence on $\alpha_V T$ from quadratic to linear. Compared to the state-of-the-art improved linear combination of Hamiltonian simulation (LCHS) method [10], our algorithm also gives better dependence on the preparation error ϵ. Especially in the time-independent case, our algorithm achieves optimal dependence on ϵ. We mention that while the two input models have different access, the complexity lower bound dependence on parameters is actually the same. (We give an explanation on this in Sec. C of SM [68]).

Additional discussions—

Inhomogeneous case: Here, we give a high-level discussion on how to generalize our algorithm to inhomogeneous ODEs which shares a strategy similar to that in Ref. [10]. By Duhamel's principle [1], the solution of a general linear ODE in Eq. (1) can be represented as

$$\vec{\mu}(T) = \mathcal{T}e^{-\int_0^T V(t)dt}\vec{\mu}(0) + \int_0^T \mathcal{T}e^{-\int_t^T V(t')dt'}\vec{b}(t)dt, \quad (6)$$

where $\mathcal{T}$ denotes the time-ordering operator. $\mathcal{T}e^{-\int_0^T V(t)dt}\vec{\mu}(0)$ is the solution for the homogeneous case and is encoded in ρ_T. Following the same procedure as in the preparation of ρ_T, each integrand $e^{-\int_t^T V(t')dt'}\vec{b}(t)$ at different time t can be encoded into a density matrix $\rho_{b,t}$. We let $|S_{b,t}\rangle$ be the purification of $\rho_{b,t}$. By discretizing the second term with m slices, one can encode $\vec{\mu}(T)$ in a density matrix ρ_{sol} that looks roughly like $\rho_{\mathrm{sol}} \propto \rho_T + T/m\sum_t \|\vec{b}(t)\|_2\rho_{b,t}$. One purification of ρ_{sol} has the form $|S_{\mathrm{sol}}\rangle \propto |0\rangle|S_T\rangle + \sqrt{T/m}\sum_t \|\vec{b}(t)\|_2^{1/2}|t\rangle|S_{b,t}\rangle$, which can then be prepared by Lindbladian simulations with the time register controlling the evolution time on an initial state $|S_{0,sol}\rangle \propto |0\rangle|\mu_0\rangle + \sqrt{T/m}\sum_t \|\vec{b}(t)\|^{1/2}|t\rangle|b_t\rangle$.

Applications: Differential equations can have vast applications in both academics and real life [1,97]. One point that we want to emphasize is that our algorithm also could have important applications in non-Hermitian physics [18–31,90]. Since the dynamics governed by non-Hermitian Hamiltonians can be directly understood as general linear ODEs, our algorithm may set the playground of using quantum computers to test various non-Hermitian effects. While the effective non-Hermitian Hamiltonian formalism exists [98], it approximates the Lindbladians well only for short-time dynamics. In comparison, what we achieved here is to exactly embed effective non-Hermitian Hamiltonian dynamics through NDME and enable long-time behavior testing. (See SM [68] Sec. D for a detailed discussion.)

Our algorithm can also be directly applied to Gibbs-state-related tasks. Given an n-qubit positive and semi-definite Hamiltonian B, its corresponding Gibbs state at the temperature $1/\beta$ is $\rho_\beta = e^{-\beta B}/Z_\beta$, with $Z_\beta = \mathrm{Tr}(e^{-\beta B})$ denoting the partition function. Using our algorithm, one can prepare the symmetric purification state $|\rho_\beta\rangle$ of ρ_β with the Grover-type speedup by noticing that $|\rho_\beta\rangle = \sqrt{2^n/Z_\beta}(e^{-\beta/2B}\otimes I_n)|\Omega\rangle$, with $|\Omega\rangle = 2^{-n/2}(\sum_i |i\rangle|i\rangle)$ and $e^{-\beta/2B}$ as the imaginary time evolution, is a special case of a linear ODE. Also, since the value of the partition function Z_β is directly encoded in ρ_T (and $|S_T\rangle$; see Sec. B4 of SM [68]), with the aid of amplitude estimation algorithms [83,84,86], our construction also provides a path to efficient partition function estimations. Note that the ability of partition function estimation is not a general feature in other quantum ODE solvers and holds only for some evolution-based methods [8–11].

*Summary and outlook—*In summary, we propose in this Letter a new quantum algorithm for linear ODEs. By combining the natural nonunitary dynamics of Lindbladians with a new technique called nondiagonal density matrix encoding, the solution to a differential equation can be encoded into a nondiagonal block of a density matrix. With the aid of advanced Lindbladian simulation algorithms and the assumption of a plausible input model, we achieve near-optimal dependence on all parameters and outperform all existing methods. We also examine its potential influence on non-Hermitian physics and Gibbs-related applications.

For future directions, since the complexity heavily depends on the results of Lindbladian simulations, our results provide motivation to further improve the results of Lindbladian simulations, especially in the time-dependent case. In this sense, it might be easier to improve our algorithm to a lower complexity than in the improved

LCHS method [10]. Beyond solving differential equations, since our algorithm achieves eigenvalue transformations beyond the framework of QSVT [38], it will also be interesting to explore whether the ideas in this Letter can be used for other eigenvalue-related problems, such as solving general systems of linear equations. We hope this Letter can invoke future studies on Lindbladian-based quantum algorithms and other quantum applications based on nondiagonal density matrix encoding.

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